Final Project:
Nexus (i.e., Reddit-like platform)

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- **Subject**: ITMD 555 Intelligent Device Apps Android
- **Professor**: James Papademas
- **Test Emulator**: Pixel 7 pro API 35
- **Language**: Java

Understanding the Idea Behind the Project

APIs:
- **Google Firebase APIs**
  - Firebase Authentication
  - Cloud Firestore
- **Web-based routing API** for cross-platform sharing (i.e., currently on localhost)

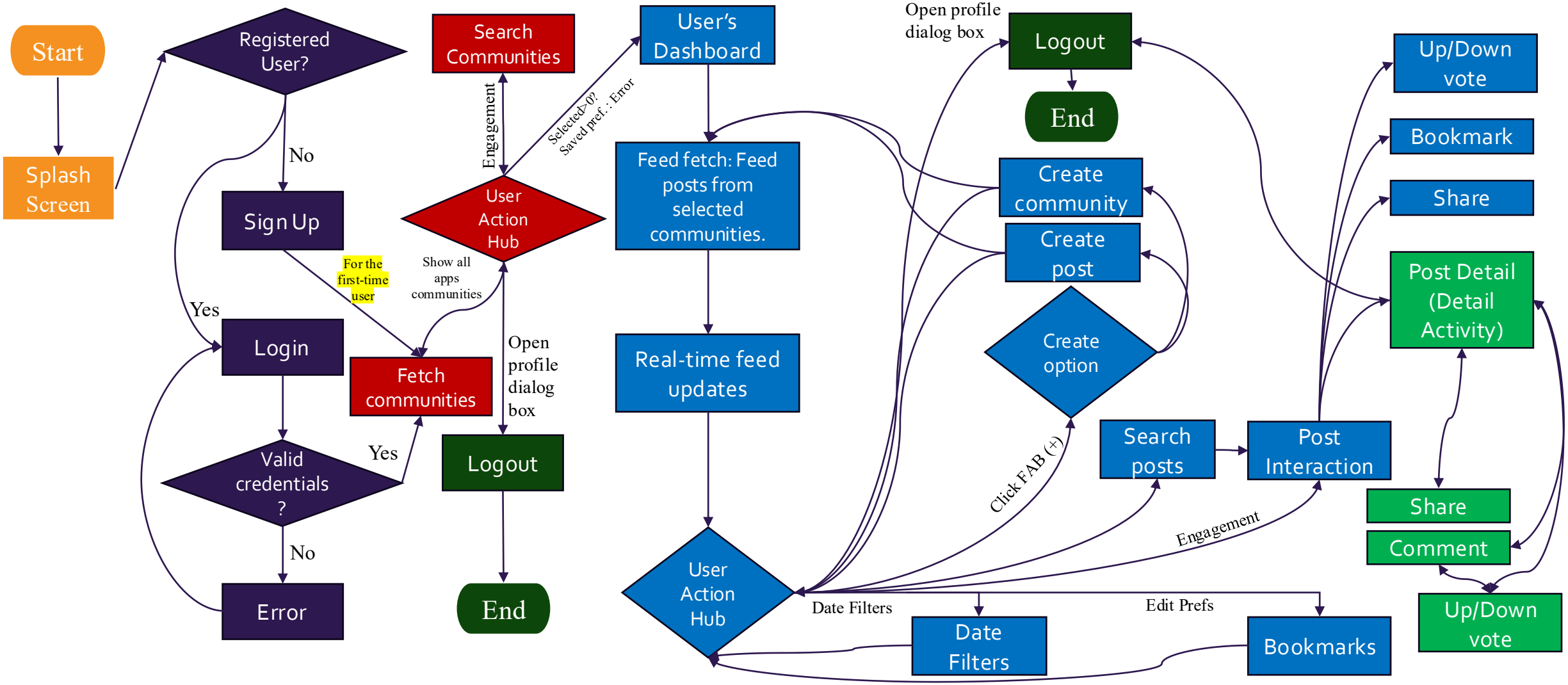
- Nexus is a **community platform, like Reddit**, that bridges fragmented social discussions to deliver high-quality information and meaningful discourse.
- Users can **join or start community pages, create posts on any community, and view posts from others**.
  - i.e., Business, Entertainment, Health, Science, Sports, Technology, Gaming, Traveling, MacBook, or any random topic, etc.

- **Groups**: Users begin by customizing their post feed through filtration (e.g., select/deselect Creative Design, Retro). This ensures the "Nexus Dashboard" is relevant from the very first login.

- **Current Features**:
  - Like, dislike (posts/comments), comments (based on threads), and share posts.
  - Bookmark posts with “Recaps” for quick access later.
  - Search content directly from the main feed.
  - Community-based feed with filtering options.
  - Share posts with other users easily.
  - Create and engage across multiple communities.
  - Backend powered by Firebase (NoSQL database).

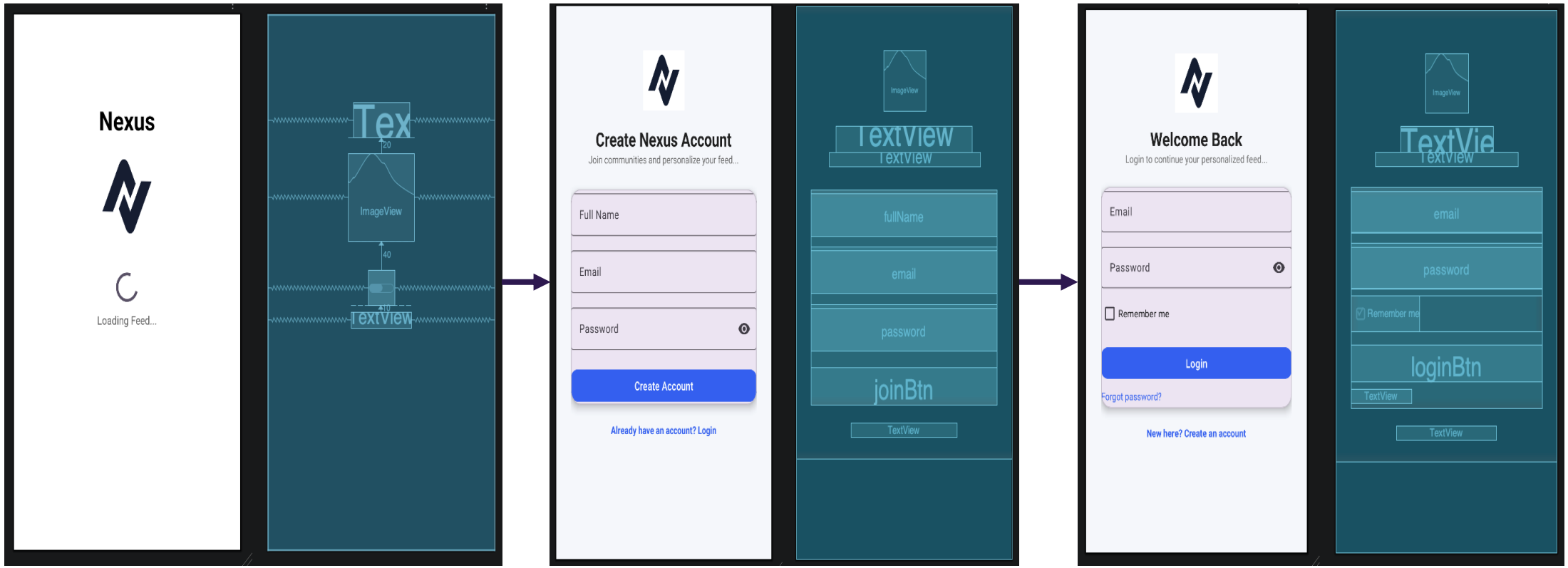
- **Firebase**:
  - Firebase Authentication (for user login & identity management)
  - Cloud Firestore (for database operations like storing users, likes, comments, etc.)

Implementation flowchart - 1



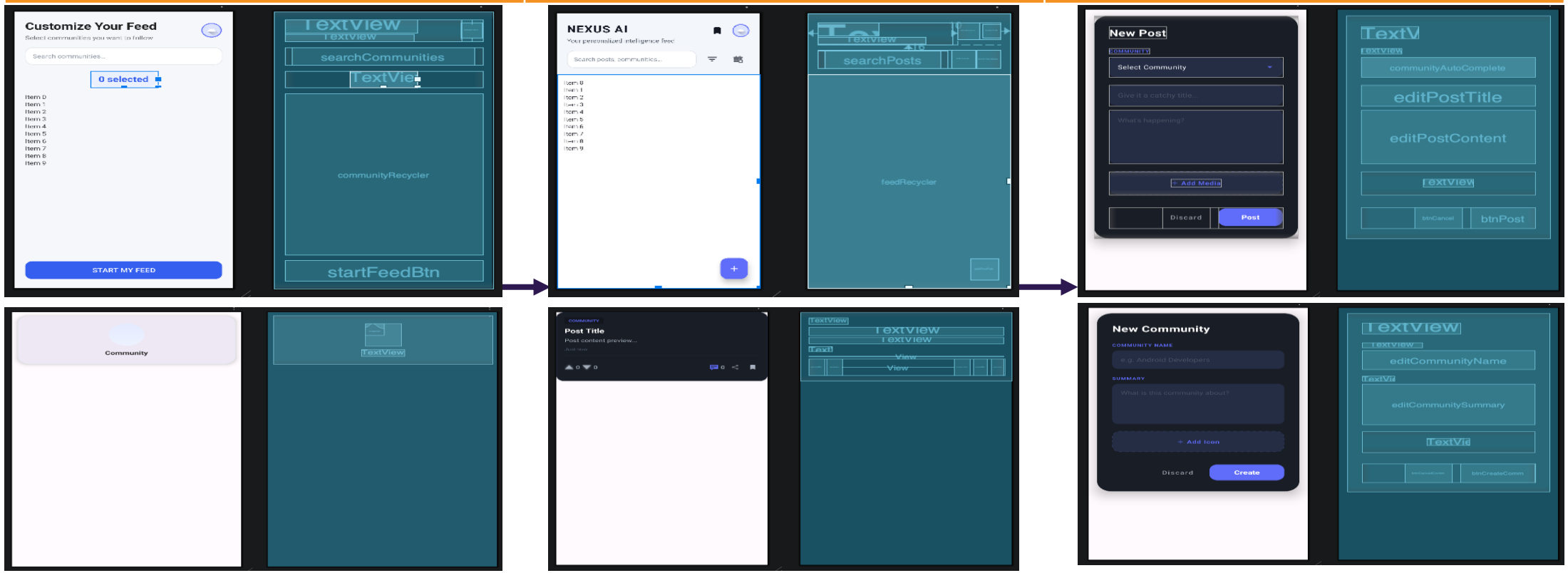
Implementation flowchart-02 (U/I part-01)

Splash Screen	Sign up	Sign in/Login



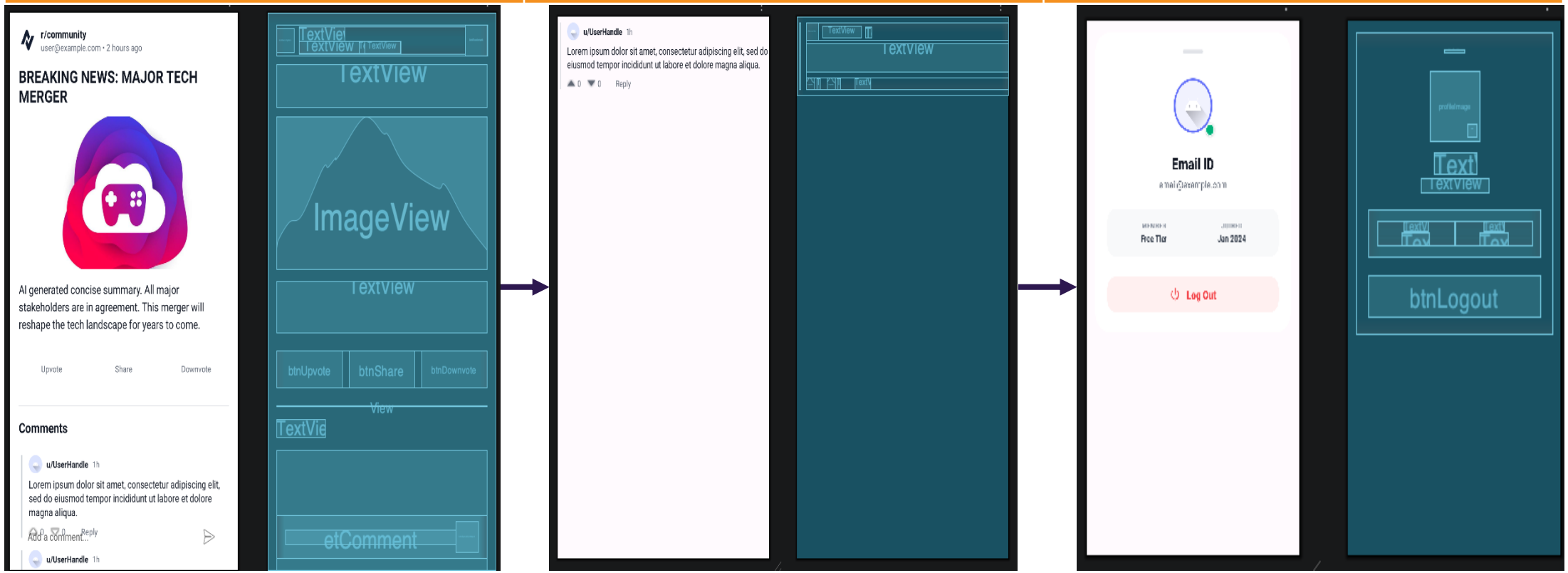
Implementation flowchart-02 (U/I part-02)

Topic selection activity	Activity Dashboard - 01	Activity Dashboard - 02



Implementation flowchart-02 (U/I part-03)

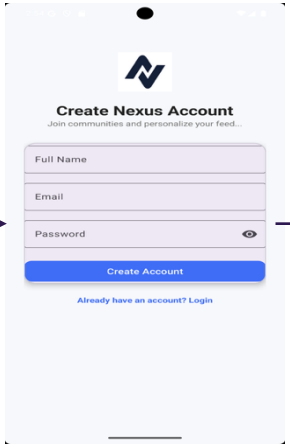
Detailed Post Activity - 01	Detailed Post Activity - 02	Logout (i.e., from any Page)



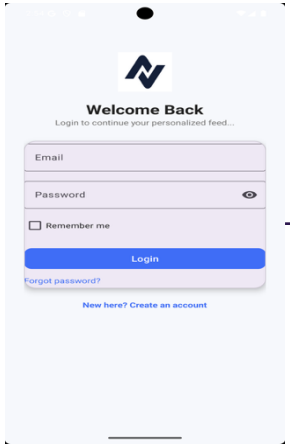
Finalized App Screenshots

Splash Screen

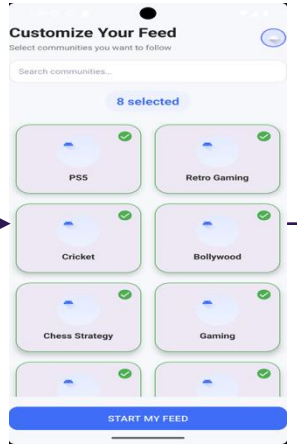
Sign up



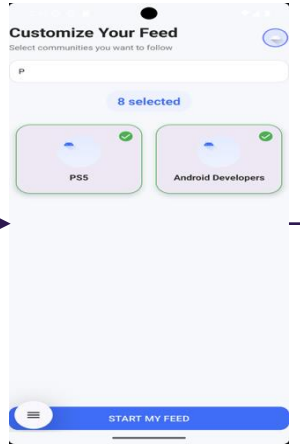
Log in



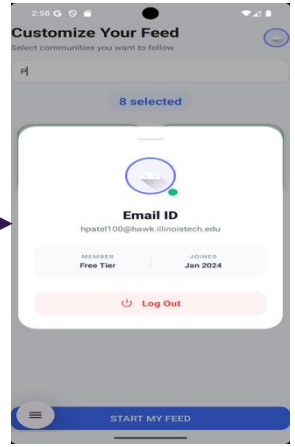
Select community



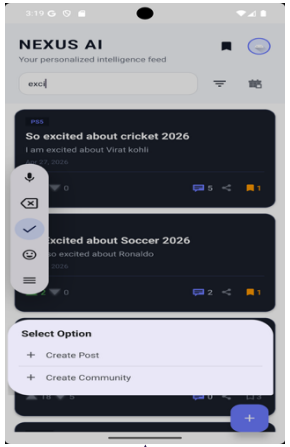
Find community



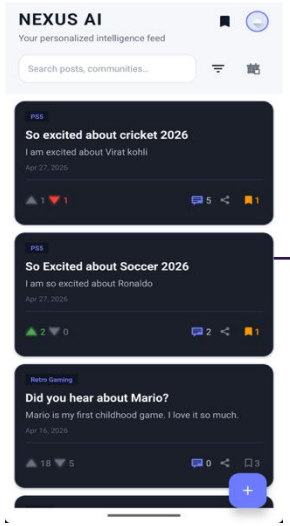
User's profile



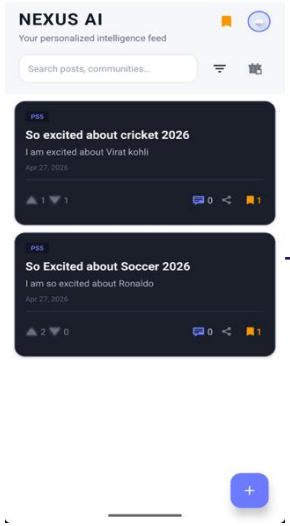
Create a new post OR community



Main Dashboard

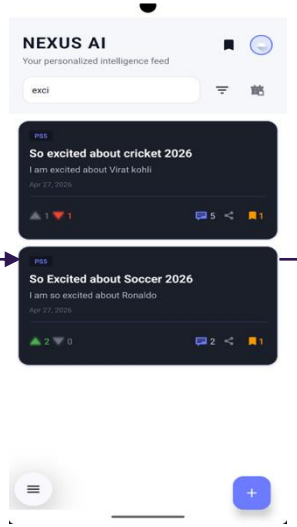


Bookmarked posts

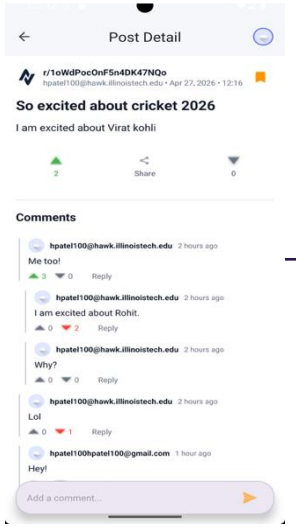


Filter by date

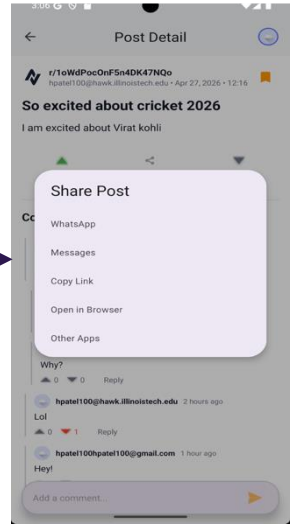
Search posts



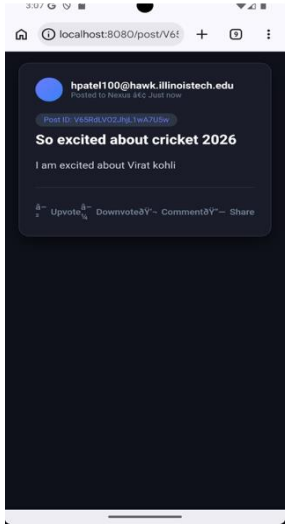
Detailed Post



Share the post

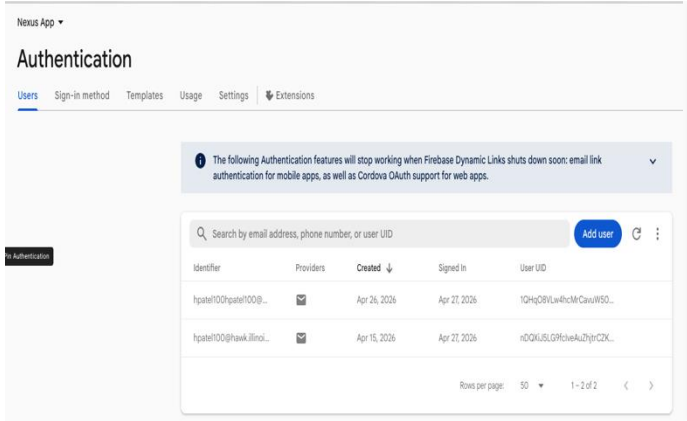


Shared post

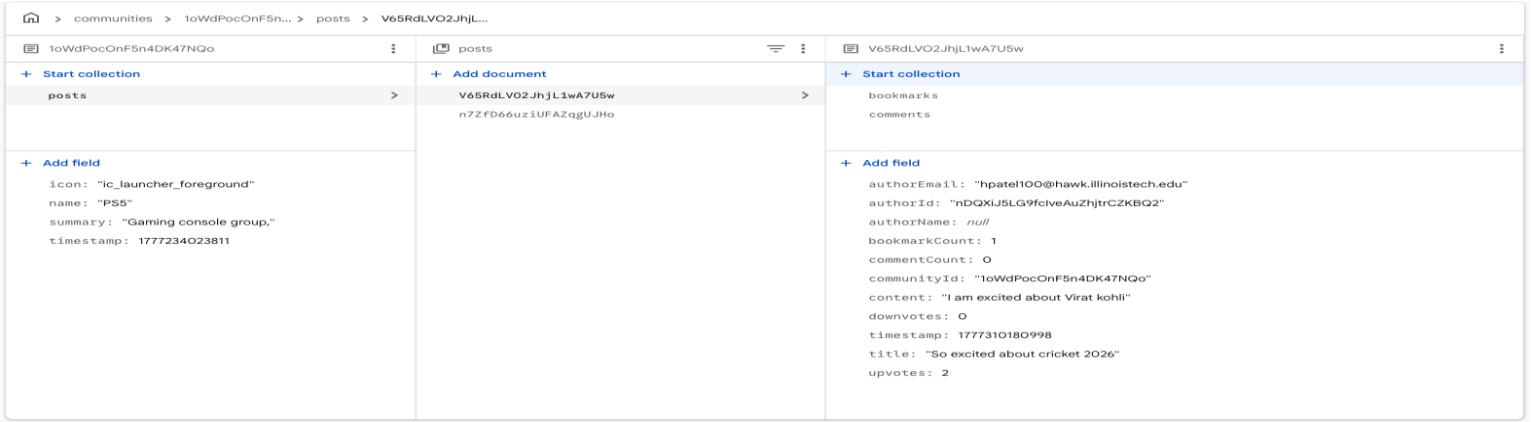


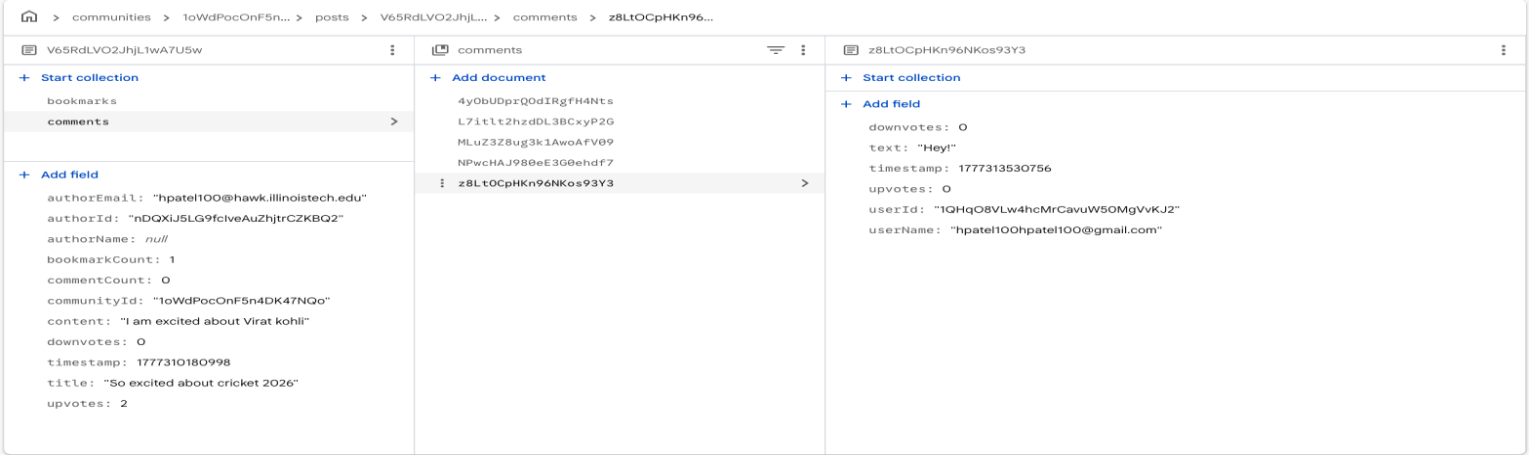
Firebase DB Implementation

User's Authentication



Communities → Posts → Comments + Bookmarks





Problem areas encountered:

- **Like/Dislike System:** Stores individual user reactions in a database for AI-based analysis and summarization.
- **Comment System:** Saves threaded comments and tracks likes on each comment.
- **Local Deployment:** Hosted on localhost for development and testing (no external server required currently).
- **U/I Design:** Focus on building and refining user-friendly interface designs.
- **Emulator Selection:** Choosing a lightweight emulator optimized for smooth performance on a MacBook (2020 model).

Future App modifications and tweak suggestions:

- **Live Deployment**: Hosting the app and collecting real user feedback and reviews.
- **Performance Optimization**: Integrating native libraries for faster and more efficient access.
- **U/I Revamp**: Designing a fresh, modern interface inspired by Reddit-style user experience.